Current Curriculum vs. Proposed Curriculum

Purpose

This document provides a two-choice comparison for HLC documentation:

- the current IT Software Developer curriculum structure
- the proposed curriculum structure

The comparison is intentionally limited to the current curriculum and the proposed curriculum. It does not include a third future-program option.

General education and shared required courses are included where they appear in the current program snapshot so the comparison reflects the full program structure instead of only the `10-152` technical course sequence.

Summary of Proposed Curriculum Changes

The proposed curriculum preserves the overall semester progression while modernizing course names, reducing overlap, and adding clearer alignment with current software, data, cloud, AI-assisted, and delivery practices.

Primary changes include:

- retaining the first-semester programming, web, and algorithmic foundations while updating course names and descriptions
- reframing SQL and advanced data work into database design and modern data modeling
- folding standalone ASP.NET exposure into the revised C# application course
- removing `10-152-125 ASP.NET Programming` as a standalone course
- retaining Docker, cloud, systems analysis, and C++ while modernizing their focus
- moving machine learning foundations earlier in the sequence
- repurposing the later advanced course space for deep learning and technical intelligence studio work
- retaining general education requirements as part of the full program structure

---

Semester 1 Comparison

Current Curriculum	Proposed Curriculum	Change Type
`10-151-101` Introduction to Security	`10-151-101` Introduction to Security	Retained; not part of current `10-152` name revision
`10-152-117` Python Programming	`10-152-117` Python Programming Foundations	Renamed and reframed
`10-152-118` HTML, CSS & JavaScript	`10-152-118` Web Development Foundations	Renamed and reframed
`10-152-119` Introduction to Algorithms	`10-152-119` Algorithmic Problem Solving	Renamed and reframed
`10-801-136` English Composition I	`10-801-136` English Composition I	General education retained
`10-804-189` Introductory Statistics	`10-804-189` Introductory Statistics	General education retained

Proposed Semester 1 `10-152` Course Descriptions

Course No.	Proposed Curriculum Name	Draft Description
`10-152-117`	Python Programming Foundations	Learners explore Python programming by designing and implementing solutions to real-world problems. Students develop foundational programming skills using variables, data types, conditionals, loops, functions, and data structures. The course emphasizes problem-solving, debugging, and code comprehension while introducing the use of AI-assisted tools to support development. Learners practice building, evaluating, and refining programs both independently and with guided assistance. By the end of the course, students are able to create, understand, and explain basic Python programs and apply structured thinking to solve problems.
`10-152-118`	Web Development Foundations	Learners develop foundational web development skills using HTML, CSS, and JavaScript to create structured, styled, and interactive web pages. The course emphasizes building multi-page websites, applying design principles for usability and readability, and implementing basic interactivity through script-based behaviors. Learners practice debugging and refining web solutions using progressive techniques and are introduced to AI-assisted tools to support development. By the end of the course, students are able to design, build, and explain functional web solutions for varied contexts and needs.
`10-152-119`	Algorithmic Problem Solving	Learners are introduced to algorithms as structured approaches to solving problems. The course focuses on analyzing problems, selecting appropriate strategies, and implementing solutions using common data structures. Students explore how different approaches affect performance through basic analysis techniques and comparison of solutions. Emphasis is placed on reasoning, debugging, and refining solutions both manually and with AI-assisted tools. By the end of the course, learners are able to evaluate, adapt, and explain algorithmic approaches used to solve problems effectively.

Semester 2 Comparison

Current Curriculum	Proposed Curriculum	Change Type
`10-152-120` SQL Programming	`10-152-120` Database Query and Design	Renamed and reframed
`10-152-121` Advanced Python Programming	`10-152-121` Advanced Python Systems	Renamed and reframed
`10-152-122` C# Programming	`10-152-122` C# Application Development	Renamed and reframed; includes related application and web concepts
`10-152-123` SQL Programming Advanced	`10-152-123` Modern Data Modeling for Systems	Replaced/reframed around broader data modeling
`10-801-198` Speech	`10-801-198` Speech	General education retained

Proposed Semester 2 `10-152` Course Descriptions

Course No.	Proposed Curriculum Name	Draft Description
`10-152-120`	Database Query and Design	Learners develop database skills by working with data models, structured queries, and relational design concepts. The course emphasizes organizing information, defining relationships, and constructing valid queries to create, retrieve, and manage data. Students practice using database tools to build tables, enforce keys, join related data, and generate useful outputs. The course is designed to support evolving platforms and workflows while strengthening the learner's ability to reason about how data is structured, accessed, and maintained in modern software systems.
`10-152-121`	Advanced Python Systems	Learners deepen their Python skills by designing more capable, maintainable, and efficient software solutions. The course emphasizes structured program design, reusable components, data handling, testing, profiling, and the use of widely adopted libraries and development tools, including structured data handling and ORM-style application workflows. Students work with larger workflows involving files, datasets, automation, and analysis while practicing debugging, documentation, and code quality

Course No.	Proposed Curriculum Name	Draft Description
		techniques. By the end of the course, learners are able to build and explain more advanced Python-based systems and adapt their approach to a range of technical problems and environments.
10-152-122	C# Application Development	Learners develop application-development skills using C# and related web application concepts to design and implement interactive software solutions. The course emphasizes core programming structures, object-oriented design, event-driven behavior, collections, file handling, exception management, and data access, while also introducing web-oriented application patterns. Students practice debugging, extending, and refining applications in ways that can adapt to changing tools and frameworks. By the end of the course, learners are able to build, explain, and improve applications that integrate both software logic and user-facing functionality.
10-152-123	Modern Data Modeling for Systems	Learners examine modern approaches to organizing, modeling, and managing data for software systems. The course emphasizes selecting appropriate data structures and storage patterns, working across relational and non-relational models, and understanding how data design affects reporting, integration, security, and scalability. Students compare alternative approaches, transform and interpret data, and practice reasoning about how information should be represented for different technical and business needs. By the end of the course, learners are able to explain and apply data-modeling choices in contemporary software environments.

Semester 3 Comparison

Current Curriculum	Proposed Curriculum	Change Type
`10-152-124` Introduction to Docker and Containers	`10-152-124` Containerized App Deployment	Renamed and reframed
`10-152-125` ASP.NET Programming	Content folded into `10-152-122`; standalone course removed	Merged/removed as standalone course
`10-152-126` Cloud for Developers	`10-152-126` Cloud Platforms for Developers	Renamed and reframed
`10-152-127` AI-Driven Systems Analysis	`10-152-127` Systems Analysis and Design	Renamed and reframed
`10-152-128` C++ Programming	`10-152-128` C++ Systems Performance Foundations	Renamed and reframed
`10-809-195 / 20-809-287` Economics option	`10-809-195 / 20-809-287` Economics option	General education/social science option retained
Not listed in current Semester 3 snapshot	`10-152-131` Machine Learning Foundations	Moved earlier into Semester 3

Proposed Semester 3 `10-152` Course Descriptions

Course No.	Proposed Curriculum Name	Draft Description
10-152-124	Containerized App Deployment	Learners develop practical deployment skills by creating, configuring, and managing containerized application environments. The course emphasizes packaging software, organizing dependencies, and supporting consistent execution across development and delivery contexts. Students practice building images, running services, and refining container-based workflows through hands-on exercises. By the end of the course, learners are able to explain how containerization supports modern software delivery and apply those concepts to evolving tools, platforms, and deployment scenarios.
10-152-126	Cloud Platforms for Developers	Learners explore cloud development practices by working with platform services, scalable application patterns, and managed infrastructure concepts. The course emphasizes selecting and using cloud resources, understanding service integration, and supporting reliable application delivery in hosted environments. Students practice working with modern developer workflows, managed tools, and data services while examining tradeoffs related to scale, availability, and maintainability. By the end of the

Course No.	Proposed Curriculum Name	Draft Description
		course, learners are able to explain and apply cloud-development patterns that can adapt to changing providers and platform ecosystems.
10-152-127	Systems Analysis and Design	Learners apply systems analysis and design methods to investigate problems, define objectives, evaluate constraints, and develop solution plans. The course emphasizes structured requirements work, workflow analysis, data and process definition, and responsible use of modern automation and AI-supported capabilities within larger systems. Students practice comparing solution paths, documenting decisions, and aligning technical choices with organizational needs. By the end of the course, learners are able to frame problems clearly, justify design decisions, and explain how systems can evolve responsibly as technologies and business requirements change.
10-152-128	C++ Systems Performance Foundations	Learners build systems-level programming foundations using C++ to explore how software behavior is shaped by data representation, control flow, and performance-aware design. The course emphasizes structured logic, input and output, data types, branching, and problem solving while introducing lower-level reasoning that supports broader understanding of runtime behavior. Students practice writing, debugging, and refining programs in ways that strengthen computational thinking across languages and platforms. By the end of the course, learners are able to explain and apply foundational systems-programming concepts in practical development contexts.
10-152-131	Machine Learning Foundations	Learners explore foundational machine learning concepts by working with common model types, sample datasets, and practical evaluation methods. The course emphasizes understanding how models are selected, trained, tested, and interpreted in support of software and decision-making tasks. Students examine expectations, compare outcomes, and evaluate effectiveness using real-world examples while also considering the role of AI-assisted tools in model-oriented workflows. By the end of the course, learners are able to explain basic machine learning processes, apply common techniques, and assess results in context.

Semester 4 Comparison

The supplied current snapshot lists Semesters 1 through 3. The Semester 4 current entries below are included from the existing curriculum comparison materials so the HLC comparison can show the full proposed program direction.

Current Curriculum	Proposed Curriculum	Change Type
`10-152-129` Agile Practices	`10-152-129` Agile Delivery and DevOps	Renamed and reframed
`10-152-130` Software Career Experience	`10-152-130` Software Career Experience	Retained and reframed as applied professional experience
`10-152-131` Machine Learning and AI	Moved earlier as `10-152-131` Machine Learning Foundations	Moved from Semester 4 to Semester 3
`10-152-132` Mobile Development	`10-152-132` Mobile Application Development	Renamed and reframed
`10-152-133` Enterprise Data Systems & API Architecture	`10-152-133` Deep Learning Frameworks and Applications	Repurposed
Not listed in current curriculum	`10-152-xxx` or `10-156-xxx` Technical Intelligence Studio	New proposed curriculum course
`10-809-199` Psychology of Human Relations	`10-809-199` Psychology of Human Relations	General education/social science course retained

Proposed Semester 4 `10-152` Course Descriptions

Course No.	Proposed Curriculum Name	Draft Description
`10-152-129`	Agile Delivery and DevOps	Learners develop team-based delivery skills by planning, building, and refining software in a collaborative project environment. The course emphasizes SDLC awareness, version control systems, review workflow, light project management and team coordination practices, and shared responsibility for software quality. Students use current collaboration and delivery tools to simulate professional development practices while adapting to changing requirements and technical constraints. By the end of the course, learners are able to contribute to structured team workflows and explain how delivery practices support reliable software outcomes.
`10-152-130`	Software Career Experience	Learners gain applied professional experience through mentored participation in workplace, field-based, simulated, service, capstone, or industry-related settings. The course emphasizes professional communication, problem solving, accountability, reflection, and the ability to connect prior technical learning to practical environments. Students document work, evaluate outcomes, and explain how their experience supports continued growth and employability. By the end of the course, learners are able to describe and demonstrate how their skills transfer into real-world software and technology contexts.
`10-152-132`	Mobile Application Development	Learners explore mobile application development by designing and implementing software for device-based and cross-platform environments. The course emphasizes user interaction, platform-aware design, structured development practices, and integration with device capabilities and external services. Students practice building, testing, and refining mobile solutions while considering usability, maintainability, and changing platform expectations. By the end of the course, learners are able to create and explain mobile application solutions that respond to real-world user and technical requirements.
`10-152-133`	Deep Learning Frameworks and Applications	Learners explore deep learning frameworks and applications by building and evaluating model-based solutions for tasks such as classification, prediction, and pattern recognition. The course emphasizes tensors, model architecture, training workflows, evaluation methods, and practical use of widely adopted development frameworks. Students compare implementation approaches, experiment with model behavior, and refine solutions based on evidence and performance goals. By the end of the course, learners are able to explain and apply deep learning workflows in ways that can adapt as tools, frameworks, and application areas evolve.
`10-152-xxx` or `10-156-xxx`	Technical Intelligence Studio	Learners investigate an emerging technology area, evaluate its relevance, and complete a bounded implementation or demonstrable use case under instructor guidance. The course combines technology monitoring, structured analysis, experimentation, communication, and reflection into a single applied studio experience. Students present findings, explain implications for software practice, and develop a personalized learning plan that supports continued growth in a changing technical landscape. By the end of the course, learners are able to analyze, implement, demonstrate, and justify technology choices in a professional context.

HLC Reading Notes

- This document compares only two choices: the current curriculum and the proposed curriculum.
- The proposed curriculum keeps the current program recognizable while updating technical content, course identity, and sequencing.
- General education courses are shown as retained where they appear in the current program structure.
- `10-152-125 ASP.NET Programming` is not retained as a standalone proposed course because the relevant exposure is folded into `10-152-122 C# Application Development`.
- `10-152-131 Machine Learning Foundations` is moved earlier so it can precede the proposed deep learning follow-on work.
- The proposed `Technical Intelligence Studio` course number remains to be finalized.